Assignment 5: Unsupervised Clustering & Dimensionality Reduction

Course: Program Elective Course-I [23CS3507]
Category: Assignments | Assignment #5 | Academic Year: 2026-27

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1. Overview & Objective:

Centroid and density clustering algorithms, and variance maximization dimensionality reduction.

2. Assignment Problems & Instructions:

1. Explain K-Means clustering algorithm steps and prove convergence to local minima.
2. Describe DBSCAN density-based clustering core points, border points, and noise.
3. Derive Principal Component Analysis PCA through eigenvalue decomposition of covariance matrix.
4. Compare Agglomerative Hierarchical clustering linkage criteria single, complete, average .

Submission Guidelines & Verification:

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